

TP 1 — Environment Setup & “Hello World”

Objectives:

- Set up development environment (VS Code, Node.js, Python, Git)
 - Create GitHub repository for course project
 - Run simple HTML “Hello World” locally
 - Commit and push to GitHub
- Outcome: Students understand client-server basics and have a working workspace.

Learning Objectives

By completing this practical, students will be able to:

- Set up a basic web development environment
- Understand the role of client and server in web development
- Create and run a simple HTML file locally
- Use Git and GitHub for version control

1. Installing the Development Environment

1.1 Visual Studio Code (VS Code)

VS Code is a lightweight and widely used code editor.

Steps:

1. Go to the official website: <https://code.visualstudio.com>
2. Download the version for your operating system (Windows / macOS / Linux)
3. Install using default settings

Recommended VS Code extensions:

- **Live Server** (for running HTML locally)
- **Prettier** (code formatting)
- **GitLens** (Git integration)

1.2 Node.js

Node.js allows you to run JavaScript outside the browser and is required for many modern tools.

Steps:

1. Visit <https://nodejs.org>
2. Download the **LTS (Long Term Support)** version
3. Install with default options

Verification:

Open a terminal and run:

```
node -v  
npm -v
```

1.3 Python

Python is often used for backend development and scripting.

Steps:

1. Visit <https://www.python.org>
2. Download Python 3.x
3. During installation, **check “Add Python to PATH”**

Verification:

```
python --version
```

1.4 Git

Git is a distributed version control system.

Steps:

1. Visit <https://git-scm.com>
2. Download and install Git
3. Use default installation settings

Verification:

```
git --version
```

2. Creating a GitHub Repository

1. Create an account at <https://github.com>
2. Click **New Repository**
3. Repository name: web-development-course
4. Choose **Public**
5. Initialize with:
 - README.md ✓
6. Click **Create Repository**

3. Local Project Setup

1. Create a local project folder:

```
mkdir web-development-course
```

```
cd web-development-course
```

2. Initialize Git:

```
git init
```

3. Connect to GitHub repository:

```
git remote add origin https://github.com/USERNAME/web-development-course.git
```

4. Creating “Hello World” HTML Page

1. Create a file called `index.html`
2. Add the following code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Hello World</title>
</head>
<body>
  <h1>Hello, World!</h1>
  <p>This is my first web page.</p>
</body>
</html>
```

5. Running HTML Locally

Option 1: Browser

- Double-click `index.html`

Option 2: Live Server (recommended)

- Right-click `index.html` in VS Code
- Select **Open with Live Server**

6. Commit and Push to GitHub

```
git add .
git commit -m "Initial setup and Hello World page"
git push -u origin main
```

Outcome of TP 1

Students:

- Understand basic client-side execution

- Have a working development environment
- Successfully use GitHub for version control